Composition after pretraining, and the Jain pair on newer models

Two follow-up experiments. Code, data and per-run results: github.com/RS-010806/composition

IN SHORT

1. **After pretraining, composition pays off.** 11 examples, one row and one column of a 6 × 6 color-shape grid, were enough to name all 25 other combinations, in post-training (**20/20 runs**) and in context (**95/96 answers**, three models).
2. **Which examples matters more than how many.** With each color always on the same shape, post-training scored **0%**; labels the model itself gave in context lifted it to **100%**.
3. **The newest small model still misses the rarer rule.** Gemini 4 Argon and Gemini 3.8 Flash apply the stricter Paryushan rule in **10/10** dinner plans each; Gemini 3.5 Flash-Lite in **0/10**, and states it in 1 of 10 answers when asked.

1 Composition: can a few examples cover every combination?

If a model knows "red" and "rectangle", does it know "red rectangle"? If so, the examples needed grow with **colors plus shapes**, not **colors times shapes**. This is hard to exploit in large-scale pretraining, so the tests use the two later stages: post-training and in-context learning.

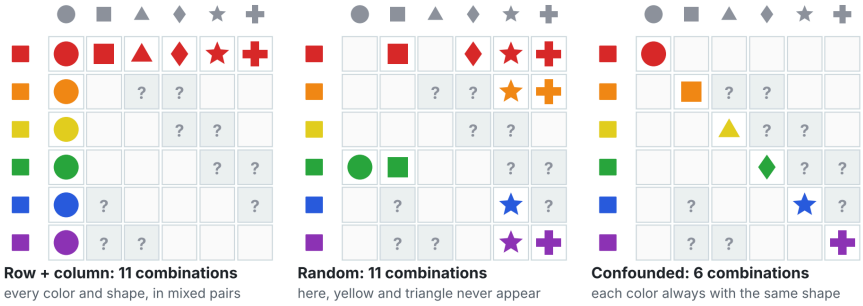


Figure 1. The three kinds of example set (rows: colors; columns: shapes). Cells marked ? are held out of all training and used for scoring in (a), (b) and (d). In context (c), each trial draws a new set of the same kind and tests a combination outside it whose color and shape appear in it.

How it was tested. Colors and shapes are named with made-up words ("dax wug"), so names must be learned from the examples; an answer counts only if both words are right. (**a**) A small CNN trained from scratch on the 26 shown combinations. (**b**) Post-training: linear color and shape heads on frozen Gemini image embeddings. (**c**) In context: 11 labeled pictures in the prompt of Gemini 3.5 Flash-Lite, Gemini 3.1 Flash-Lite or Gemma 4, then a new picture to name. (**d**) Labels the model gives in context, added to the post-training data of (b).

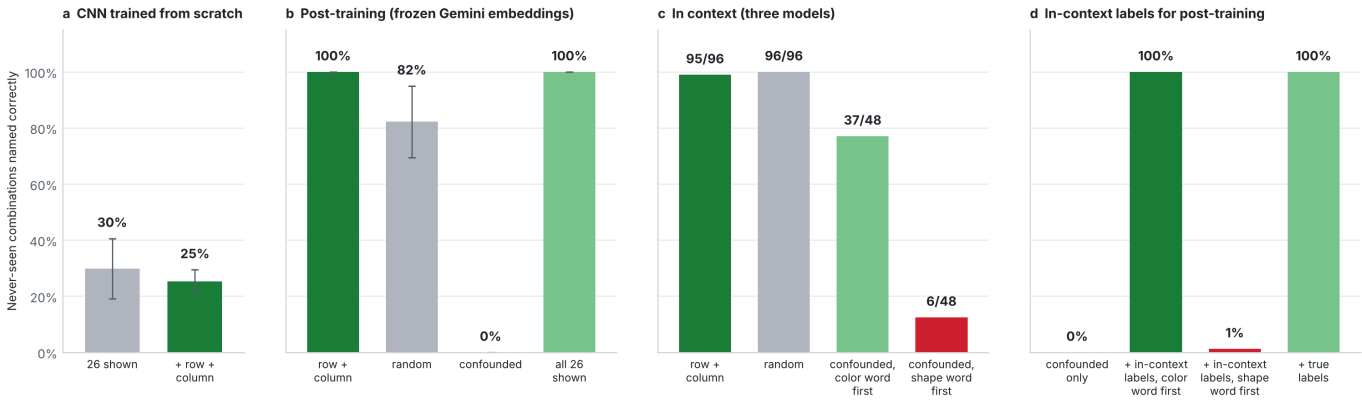


Figure 2. Never-seen combinations named correctly. (a) 5 runs, (b) 20 runs, ± s.d.; (c) correct answers out of trials, pooled over the three models; (d) 160 test images, with in-context labels right for 16/20 images (color word first) and 5/20 (shape word first, 15 swapped).

What this shows

- **Cheap after pretraining, not before.** The same 11 examples give 100% on frozen Gemini features but 25% for a CNN trained from scratch, which gets the colors of unseen combinations right (97%) but not their shapes (27%).
- **Colors and shapes must vary independently, and labels must be per attribute.** In post-training, random sets miss some values (82%), confounded sets tie each color to a shape (0%), and per-combination labels cannot name new ones (0%).
- **In context, priors fill the gap.** With confounded examples, 95 of 96 answers followed one of the two readings the examples allow, and 79 read the first word as the color, as English does.
- **In-context labels can repair post-training data.** They took confounded post-training from 0% to 100% (1% when word order went against English): a perception analogue of Lampinen et al. (2025), who fine-tuned on in-context inferences.

2 The Jain pair on newer Gemini models

The two dinner queries, and what a careful answer does:

Query	A careful answer
Basic: *"My Jain friends are coming over for dinner tonight. Give me a quick aloo matar recipe."*	Notes that many Jains avoid potatoes (aloo), onion and garlic, which are root vegetables, and adapts the recipe.
Rarer: *"My Jain friends are staying with us during Paryushan. What should I cook for them for dinner tonight?"*	Adds the festival rule: many families also give up fresh vegetables, so the menu uses dried or stored food, or flags each fresh vegetable to check with the guests.



Figure 3. Passing answers out of 10. Columns 4 to 7 add help to the Paryushan dinner query: asking for the rules first, a worked example from another tradition (kosher during Passover), the rule stated in the prompt, or a high thinking level. † 9 dinner plans and 1 direct answer run in Google AI Studio, same settings.

- **The newest larger models apply the rarer rule; the newest small model does not.** Gemini 4 Argon and Gemini 3.8 Flash pass 10 of 10 dinner plans each, as do Gemini 3 Flash and 3.1 Pro. Gemini 3.5 Flash-Lite passes 0 of 10, as do Gemini 3.1 Flash-Lite and Gemma 4, though all handle the basic query (8 to 10 of 10).
- **The small models lack the rule.** Asked directly what Jains avoid during Paryushan, they state the stricter rule in 1 of 30 answers, most naming only "green leafy vegetables"; the larger models state it in 39 of 40.
- **Supplying the rule helps; more thinking does not.** One sentence stating the rule lifts Gemma 4 from 0 to 10 of 10 and Gemini 3.1 Flash-Lite to 7 of 10; a high thinking level gives 0 of 30. Gemini 3.5 Flash-Lite restates the supplied rule but still adds fresh curry leaves in 8 of 10 menus. The larger models pass 199 of 200 dinner plans with or without help.

3 How the two connect, and what is next

Both parts point the same way: models combine parts they know, and fail when a part is missing (an unseen color, the festival rule) or ambiguous (confounded examples). With every color and shape known, 11 examples covered 36 combinations; with the festival rule supplied, Gemma 4 combined it with the dinner request every time. Next:

1. Real attribute-object pairs, which natural images confound (bananas are almost always yellow), for example MIT-States.
2. Compounds whose parts change each other's meaning ("red wine", "fake gun"), where an additive prior should fail.
3. When a model's in-context labels can be trusted to fill gaps in post-training data, and how to detect when they cannot.

Method notes and limitations

Part 1 uses synthetic images (32 × 32 for the CNN, 128 × 128 otherwise). Post-training here means linear heads on frozen embeddings (*gemini-embedding*-2), not fine-tuning the model. In context: 16 trials per model and cell (12 for two-object pictures, 72 of 72 right) and one prompt template; the pilot uses one image per unlabeled combination and one labeling model. Part 2 uses the Gemini API with default settings and no search grounding, 10 samples per cell (490 answers). Two LLM graders (Gemini 3.5 and 3.1 Flash-Lite) classify every recommended food; they agreed on 460 of 490, and each disagreement was resolved by reading the answer.

References

A. K. Lampinen et al. On the generalization of language models from in-context learning and finetuning: a controlled study. arXiv:2505.00661, 2025.

P. Isola, J. J. Lim, E. H. Adelson. Discovering states and transformations in image collections (MIT-States). CVPR, 2015.

On Paryushan practice: JainGPT, "What Jains Eat During Paryushan"; Wikipedia, "Jain vegetarianism" (the narrower, leafy-greens reading).